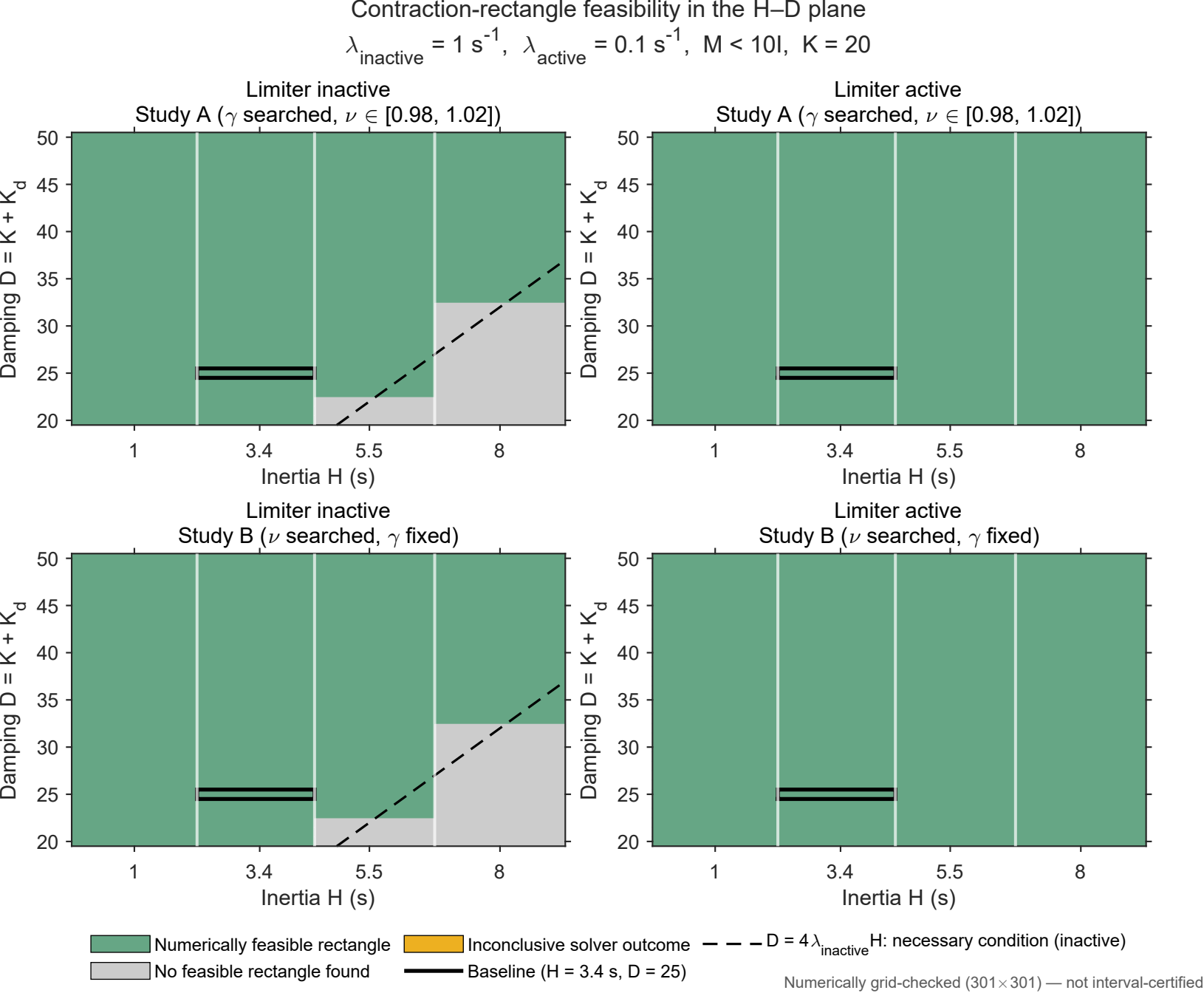
Contraction-rectangle feasibility in the H–D plane
$\lambda_{inactive}$ = 1 s^{-1}, λ_{active} = 0.1 s^{-1}, M < 10I, K = 20
Limiter inactive
Study A (γ searched, $\nu \in$ [0.98, 1.02])
Limiter active
Study A (γ searched, $\nu \in$ [0.98, 1.02])
Limiter inactive
Study B (ν searched, γ fixed)
Limiter active
Study B (ν searched, γ fixed)
Damping D = K + K_d
Inertia H (s)
50
45
40
35
30
25
20
1
3.4
5.5
8
Numerically feasible rectangle
No feasible rectangle found
Inconclusive solver outcome
Baseline (H = 3.4 s, D = 25)
D = 4 $\lambda_{inactive}$H: necessary condition (inactive)
Numerically grid-checked (301×301) — not interval-certified